

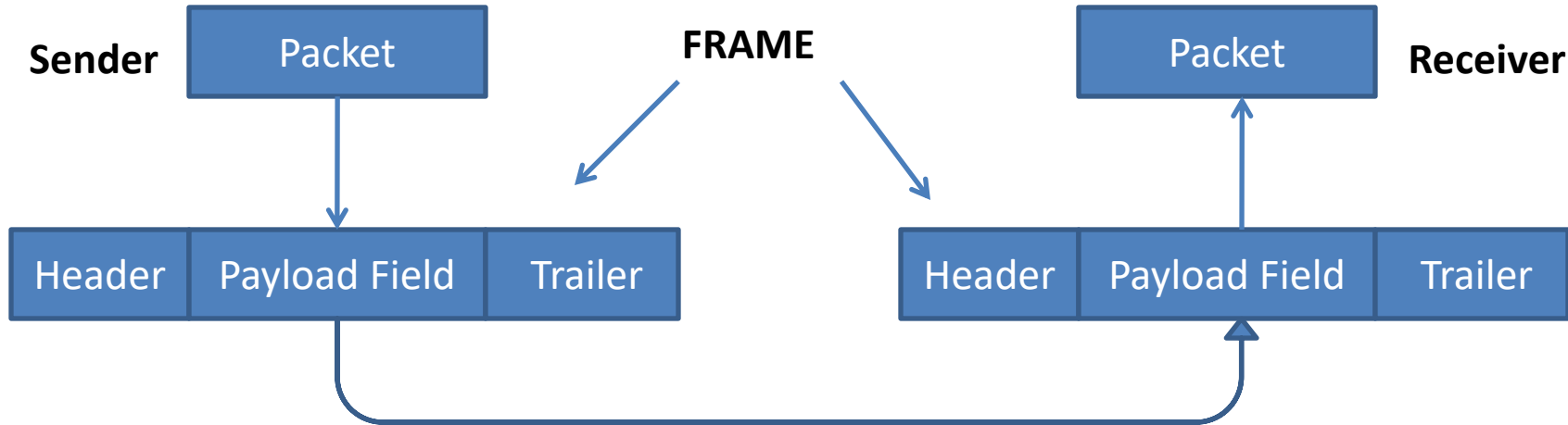
Data Link Layer:

The Data Link layer is the second layer in the OSI (Open Systems Interconnection) model. It is responsible for node-to-node data transfer and handles error detection and correction from the physical layer. The Data Link layer ensures that data transferred between adjacent network nodes is error-free and efficient.

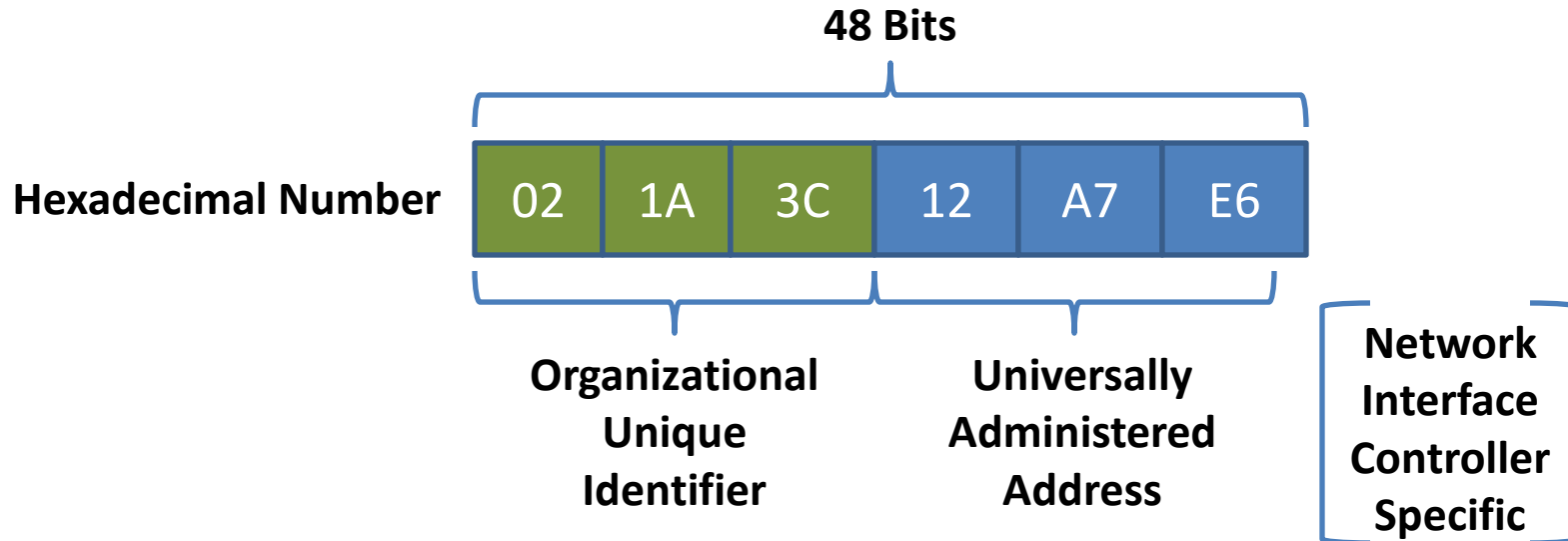
Key Functions of the Data Link Layer

- Framing
- Physical Addressing
- Error Detection and Correction
- Flow Control
- Access Control
- Link Establishment and Termination

Framing: The Data Link layer encapsulates raw bits from the Physical layer into frames. A frame is a structured package of data that includes not only the **Payload (actual data)** but also control information such as source and destination MAC addresses and protocol-specific details (in Header) and CRD and error-checking codes (In Trailer).



Physical Addressing: Unlike the Network layer, which uses logical addressing (like IP addresses), the Data Link layer uses physical addressing to identify devices on a local network. Each device's network interface card (NIC) has a unique MAC (Media Access Control) address used for this purpose.



Error Detection and Correction: The Data Link layer provides mechanisms to detect and correct errors that may occur in the Physical layer. It uses techniques like cyclic redundancy check (CRC) to detect errors in transmitted frames. If errors are detected, the frames can be retransmitted.

There are two Types of Error in Data Forwarding:

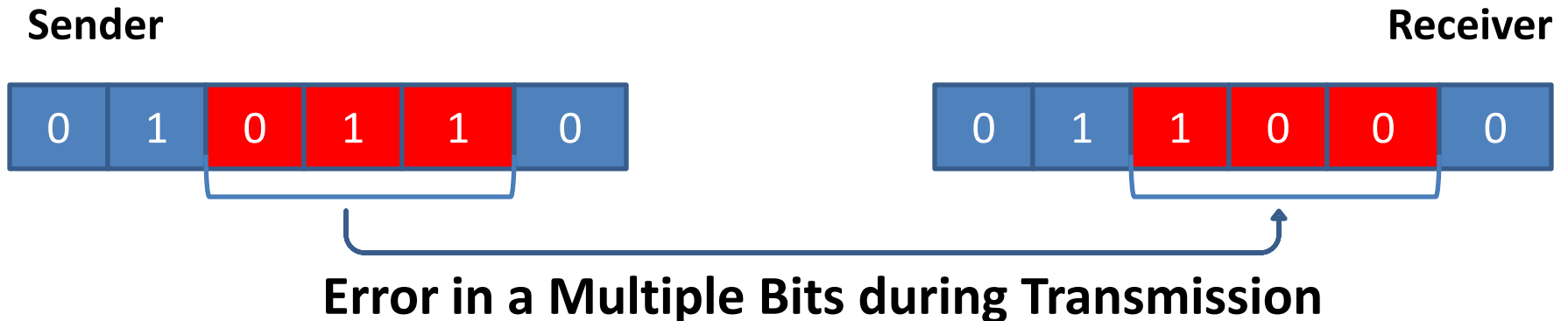
1. Bit Error (Error in a Single Bit)
2. Burst Error (Error in multiple Bit)

Two Types of Error in Data Forwarding: Data Link Layer

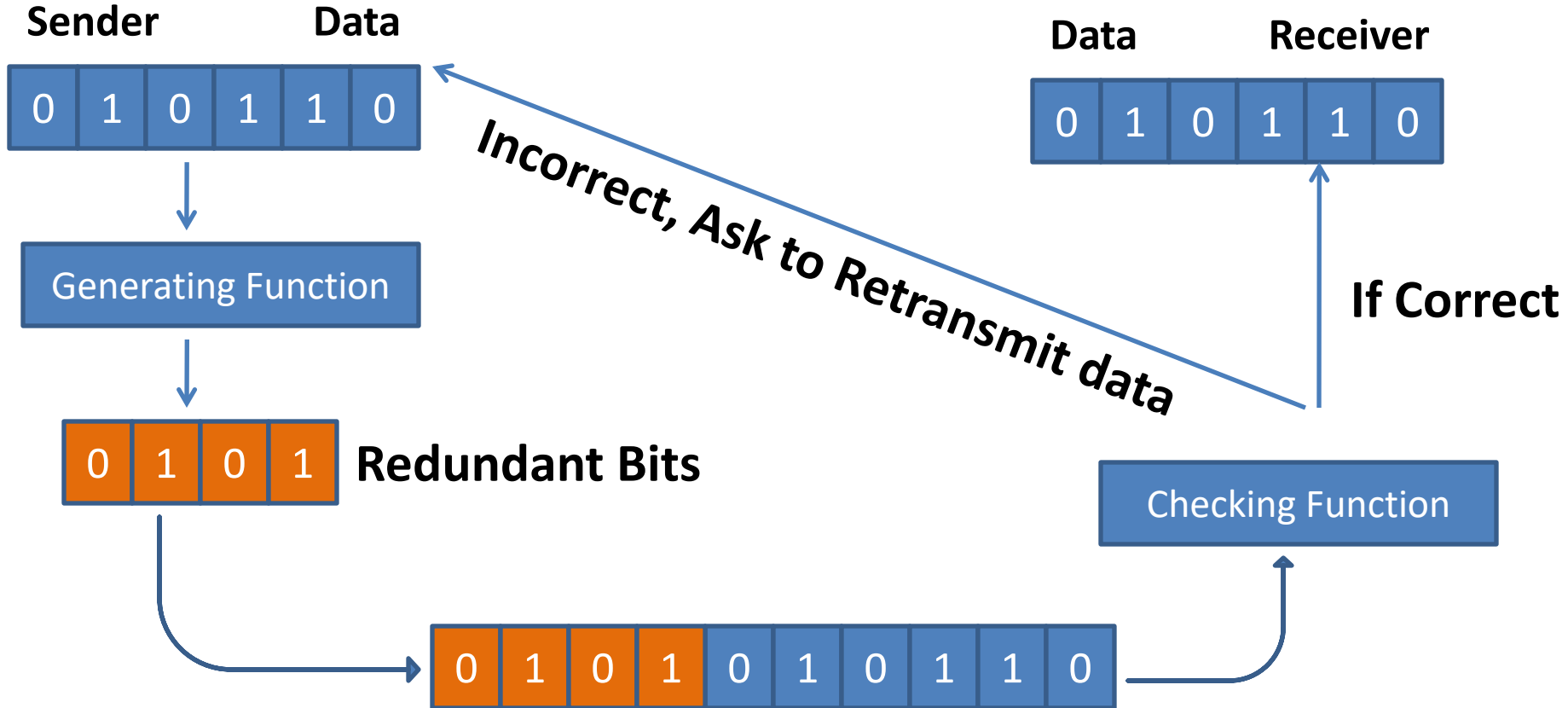
Bit Error:



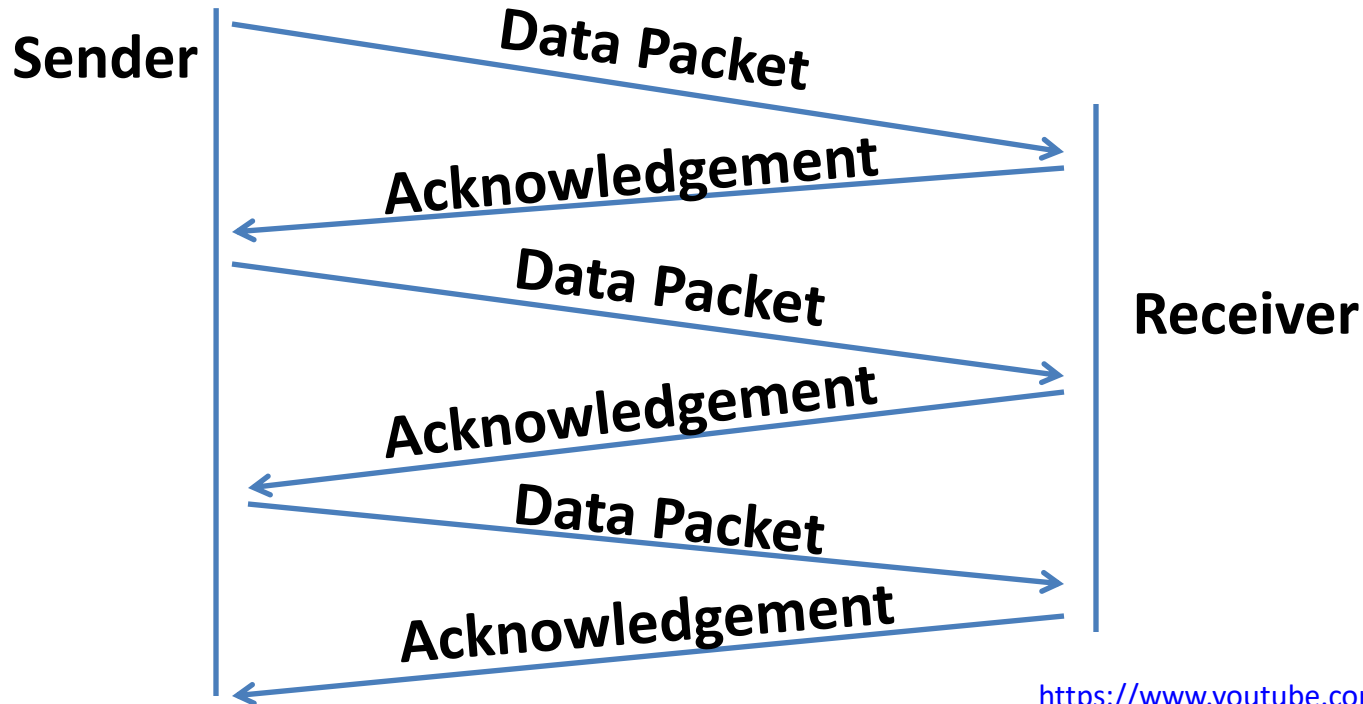
Burst Error:



Cyclic Redundancy Check:



Flow Control: Once the path is determined, the network layer forwards the packets from one network segment to another, eventually reaching the destination. This involves handling the packets at each hop along the route.



Sub layers of the Data Link Layer:

Logical Link Control (LLC):

The LLC Sublayer manages communication between the Data Link layer and the Network layer. It handles error checking and frame synchronization. The LLC Sublayer provides a way to multiplex multiple network protocols over the same physical network and controls frame synchronization and error checking. Some Functions of LLC:

- Multiplexing and Demultiplexing**
- Error Detection**
- Flow Control**
- Frame Sequencing**

Sub layers of the Data Link Layer:

Media Access Control (MAC):

The MAC sublayer is responsible for controlling how devices on a network gain access to the medium and permission to transmit data. It handles the physical addressing (MAC addresses) and defines how data is placed on and retrieved from the physical medium. Functions of MAC Sublayer:

- Physical Addressing**
- Access Control (how multiple devices share same physical medium)**
- Medium Access Control Protocols**
- Flow Control**

Physical Layer:

The Physical layer is the first and lowest layer in the OSI model. It is responsible for the actual transmission and reception of raw bit streams over a physical medium. This layer defines the hardware elements involved in the data transmission process, including cables, switches, network interface cards, and other physical components.

